Copernicus Global Land Operations – Lot I "Vegetation and Energy"

Framework Service Contract N° 945120 – IPR – 2023 (JRC)

PRODUCT USER MANUAL

LAND SURFACE TEMPERATURE
VERSION 3.0

ISSUE I1.00

Organization name of lead contractor for this deliverable: IPMA

Book Captain: Emanuel Dutra

Contributing Authors: Sofia Ermida
Isabel Trigo
Francisco Lopes

© CGLOPS Lot1 consortium

Dissemination Level		
PU	Public	X
PP	Restricted to other programme participants (including the Commission Services)	
RE	Restricted to a group specified by the consortium (including the Commission Services)	
CO	Confidential, only for members of the consortium (including the Commission Services)	

© CGLOPS Lot1 consortium

Document Release Sheet

Book captain:	Emanuel Dutra (IPMA)	Sign: Emanuel Dutra Date 14.04.2026
Approval:	Roselyne Lacaze (HYGEOS)	Sign Roselyne Lacaze Date 14.04.2026
Endorsement:	Nadine Gobron (JRC)	Sign Date
Distribution:	Public	

© CGLOPS Lot1 consortium

Change Record

Date	Page(s)	Description of Change	Release
14.04.2026	All	First Release	I1.00

© CGLOPS Lot1 consortium

TABLE OF CONTENTS

Executive Summary *11*

1. Background of the document *12*

1.1 Scope and Objectives 12

1.2 Content of the Document 12

1.3 Related Documents 12

1.3.1 Applicable documents 12

1.3.2 Input 12

1.3.3 External documents 13

2. Algorithm *14*

2.1 Overview 14

2.2 Generalized Split-Window Algorithm 14

2.3 Data Fusion 15

2.4 Known Limitations 17

2.5 Differences with previous version 17

3. Product Description *18*

3.1 File naming 18

3.2 File Format 18

3.3 Product Content 19

3.4 Product Characteristics 24

3.4.1 Projection & Grid Information 24

3.4.2 Temporal Information 24

3.5 Data Policies 25

3.6 Access and Contacts 25

4. Validation *27*

4.1 Procedure 27

4.2 Main Results 27

5. Product Usage *28*

5.1 Typical use 28

© CGLOPS Lot1 consortium

5.2 Advanced usage 28
Land surface processes and agriculture 28
5.2.1 Hydrology and water cycle 28
Urban Climate 29
Thermal anomaly monitoring 29
Climate studies 29
Model-related applications 30
6. References 31
7. References 34
8. Annex 1: Requirements 35
8.1 Specific technical details and requirements 35
8.2 Further requirements 35
8.2.1 Output product composition 35
8.2.2 Data structure 35
8.2.3 Data format 36
8.2.4 Uncertainties and validation 36
8.2.5 Input data 36
8.2.6 Product delivery 37
9. Annex 2: Differences with previous versions 38

© CGLOPS Lot1 consortium

List of Figures

Figure 1: Areas covered by GEO satellites. From left to right: GOES, MSG 0°, MSG IODC and Himawari. 15

Figure 2: Flow diagram of the generation of LST GLOBE product. 16

Figure 3: Satellite / View Zenith Angles for the four missions 16

Figure 4 : Pixel average resolution as function of latitude for GOES16 (red), HMWR (dashed magenta), MSG (blue), CLMS LST previous version 2 (black) and CLMS LST version 3 (grey). For the Geostationary, it is assumed the central longitude of each platform. 24

Figure 5: Illustration of the direction and duration of scanning of GEO satellites for the *t* UTC data. 25

© CGLOPS Lot1 consortium

List of Tables

Table 1 : Satellite channels used for LST estimation using the GSW algorithm. 15

Table 2: Explanation of the version numbering scheme and recommendations for handling version updates 18

Table 3: Value encoding for the data variables 19

Table 4: LST Quality Flags 20

Table 5: File-level metadata attributes of LST products. 21

Table 6: Metadata attributes for the main LST data variable 22

Table 7: Description of netCDF attributes for latitude and longitude coordinate variables 23

Table 8: Description of netCDF attributes for time coordinate variable 23

Table 9: Description of netCDF attributes for the grid mapping variable 23

Table 10: CGLOPS LST product specifications. 35

Table 11: Differences between product versions. 38

© CGLOPS Lot1 consortium

List of Acronyms

AD	Applicable Document
ABI	Advanced Baseline Imager
AHI	Advanced Himawari Imager
ATBD	Algorithm Theoretical Basis document
CEOS	Committee on Earth Observation Satellites
CF	NetCDF Climate and Forecast (CF) metadata conventions
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
COG	Cloud optimized geotiff
ECV	Essential Climate Variable
EPSG	European Petroleum Survey Group
ERA5	ECMWF Re-analysis version 5
ESA	European Space Agency
EU	European Union
EUMETCast	EUMETSAT's broadcast system for environmental data
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
GBOV	Ground-Based Observations for Validation of Copernicus Global Land Products Copernicus Ground-Based Observations for Validation
GCOS	Global Climate Observing System
GEO	Geostationary satellites
GOES	Geostationary Operational Environmental Satellite
GSW	Generalized Split-Window
HMWR	Himawari
IGBP	International Geosphere-Biosphere Program
INSPIRE	Infrastructure for Spatial Information in the European Community
IODC	Indian Ocean Data Coverage
IPMA	Instituto Português do Mar e da Atmosfera
JRC	Joint Research Center
LPV	Land Product Validation
LSA-SAF	EUMETSAT Satellite Application Facility on Land Surface Analysis
LST	Land Surface Temperature
MAE	Mean Absolute Error
MODIS	Moderate Resolution Imaging Spectroradiometer
MSG	Meteosat Second Generation
NetCDF	Network Common Data Form
NRT	Near Real Time
NWC SAF	EUMETSAT SAF on Nowcasting and Very Short-Range Forecasting

PQAD	Product Quality Assurance Document
PUM	Product User Manual
Q_FLAGS	Quality flag
RMSE	Root Mean Square Error
SAF	Satellite Application Facility
SCE	Snow Cover Extent
SEVIRI	Spinning Enhanced Visible Infra-Red Imager
SLSTR	Sea and Land Surface Thermal Radiometer on ESA Sentinel-3 mission
SM	Soil Moisture
SZA	Satellite or Viewing Zenith Angle
TIR	Thermal Infrared
TOA	Top-of-Atmosphere
UTC	Coordinated Universal Time
VR	Validation Report
WGS	World Geodetic System

© CGLOPS Lot1 consortium

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery take place in a timely manner and are complemented by the constitution of long-term time series.

The Product User Manual (PUM) is dedicated to CLMS users giving an overview of the product characteristics, in terms of algorithm, technical properties, and main validation results.

The present document describes the Land Surface Temperature (LST) product V3.0 retrieved from a constellation of geostationary (GEO) satellite missions: Meteosat Second Generation (MSG) 0º and Indian Ocean Data Coverage (IODC) missions, Geostationary Operational Environmental Satellite (GOES) and Himawari. These platforms are operated by different satellite agencies, but the data are available in near-real time (NRT) via the European Organization for the Exploitation of Meteorological Satellites (EUMETSAT) dissemination system, EUMETCast. The product is calculated globally every hour and made available to the users in near-real time with a timeliness of 4 hours and with a spatial resolution of 0.025°.

© CGLOPS Lot1 consortium

1. BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The Product User Manual (PUM) is the primary document that users have to read before handling the products.

It gives an overview of the product characteristics, in terms of algorithm, technical characteristics, and main validation results.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 gives a description of the retrieval algorithm;
- Chapter 3 describes the main characteristics of the product;
- Chapter 4 shows the main results on validation.

The requirements are recalled in Annex 1, and an overview of the product evolution and differences with previous version in Annex 2.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_ATBD_LST-V3.0	Algorithm Theoretical Basis Document of the Land Surface Temperature Version 3.0
CGLOPS1_VR_LST-LST10-V3.0	Validation Report the Land Surface Temperature products family Version 3.0

Available at https://land.copernicus.eu/en/technical-library

1.3.3 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs requirements.

Available online at: https://library.wmo.int/

© CGLOPS Lot1 consortium

2. ALGORITHM

LST is estimated from Top-of-Atmosphere (TOA) brightness temperatures of atmospheric window channels within the infrared range. The used split-window methodology makes use of two adjacent window channels within the thermal infrared range (T_{TIR1} and T_{TIR2}), where LST is expressed as a regression function of TOA brightness temperatures. To minimize LST uncertainties, the algorithm is trained for different classes of satellite viewing angle and atmosphere water vapour content.

In the current version of CGLOPS LST, the global LST is derived using retrievals from the Advanced Himawari Imager (AHI) onboard Himawari-8 and Himawari-9, the Advanced Baseline Imager (ABI) onboard GOES-16 and GOES-19, and the Spinning Enhanced Visible Infra-Red Imager (SEVIRI) onboard the Meteosat Second Generation (MSG) platforms in charge of the 0º and Indian Ocean Data Coverage (IODC) missions. The product includes data since January 1st 2018, ensuring coverage from all the above-mentioned sensors

2.1 OVERVIEW

The LST product is generated every hour, through the following steps:

Step 1: Generation of LST product for each GEO satellite disk.

Step 1.1: Gathering of input data in the same resolution and projection as the satellite imagery;

Step 1.2: LST production is performed for all land pixels in clear sky conditions;

Step 1.2.1: the LST is calculated with the Generalized Split Window (GSW) algorithm; where surface emissivity is first estimated considering the Land Cover, Fraction of Vegetation Cover and Snow Cover of each pixel. Clear sky conditions are identified via the cloud mask produced by the NWC SAF software.

Step 1.2.2: the uncertainties associated to the LST retrievals are calculated and set up as quality flag;

Step 2: The LST (and the additional layers) for the Himawari and GOES disks is merged with the LSA-SAF LST from both MSGs (0° and IODC missions) to obtain a global product.

2.2 GENERALIZED SPLIT-WINDOW ALGORITHM

The GSW algorithm provides estimation of LST using measurements from two adjacent channels within the infrared atmospheric window, T_{TIR1} and T_{TIR2} (Table 1:).

All inputs are re-gridded (both in time and space, when needed) for each pixel in the original satellite geostationary projection, so that all algorithm inputs are in a common grid. The following inputs are required for GSW: (i) land cover database, (ii) fraction vegetation cover,

(iii) look-up-tables for bare soil and vegetation emissivities; (iv) snow cover fraction; (v) cloud mask and (vi) total column water vapour. A detail list of these inputs is given in the product ATDB along with a list of changes in input data used since 2018 to near real time (see Annex 1 of CGLOPS1_ATBD_LST-V3.0)

Table 1 : Satellite channels used for LST estimation using the GSW algorithm.

	MSG/SEVIRI	GOES/ABI	Himawari/AHI
TIR1	10.0 – 11.5 µm	10.8 – 11.6µm	10.8 – 11.6 µm
TIR2	11.2 – 12.8 µm	11.8 – 12.8µm	11.7 - 13.1 µm

2.3 DATA FUSION

LST corresponds to instantaneous fields produced every 15 minutes in the case of SEVIRI, and hourly in the case of GOES and Himawari. The three individual fields are then re-projected and merged onto a 0.025º regular grid. The individual timeslots selected to integrate the global product are T=0 for all sensors. However, discrepancies may arise from the different configurations of the scan: SEVIRI's full-disk scan takes about 12 minutes and is performed from south to north, while ABI/AHI's scan can take between 5 to 10 minutes, depending on the mode, and is performed from north to south. Figure 1 illustrates the Earth disks observed by each GEO satellite in the original projection.

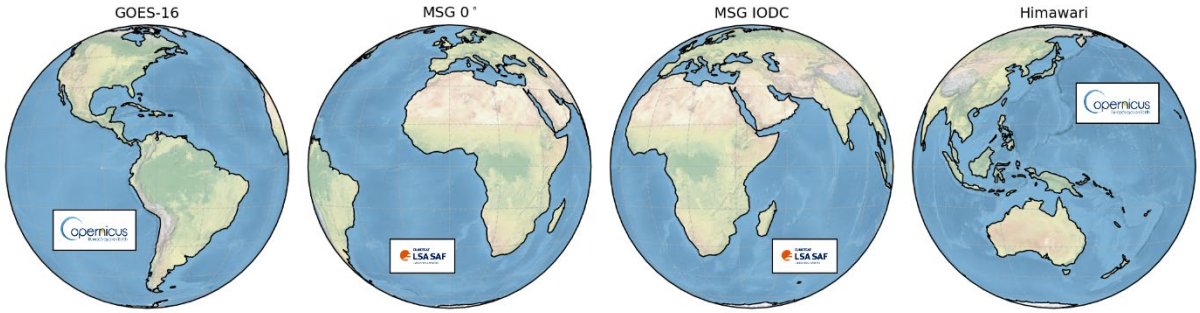


Figure 1: Areas covered by GEO satellites. From left to right: GOES, MSG 0°, MSG IODC and Himawari.

LST data fusion consists of LST retrievals from the different GEO satellites merged into a single field, thus yielding a global product (see Figure 2). The final LST product is obtained by averaging all the LST values from pixels within each grid box (regular in latitude and longitude), for SZAs up to 70º for all sensors. In this context, all the averages are weighted considering the respective viewing angle, so that the final product is less influenced by lower quality retrievals (i.e., those with greater SZA, see Figure 3).

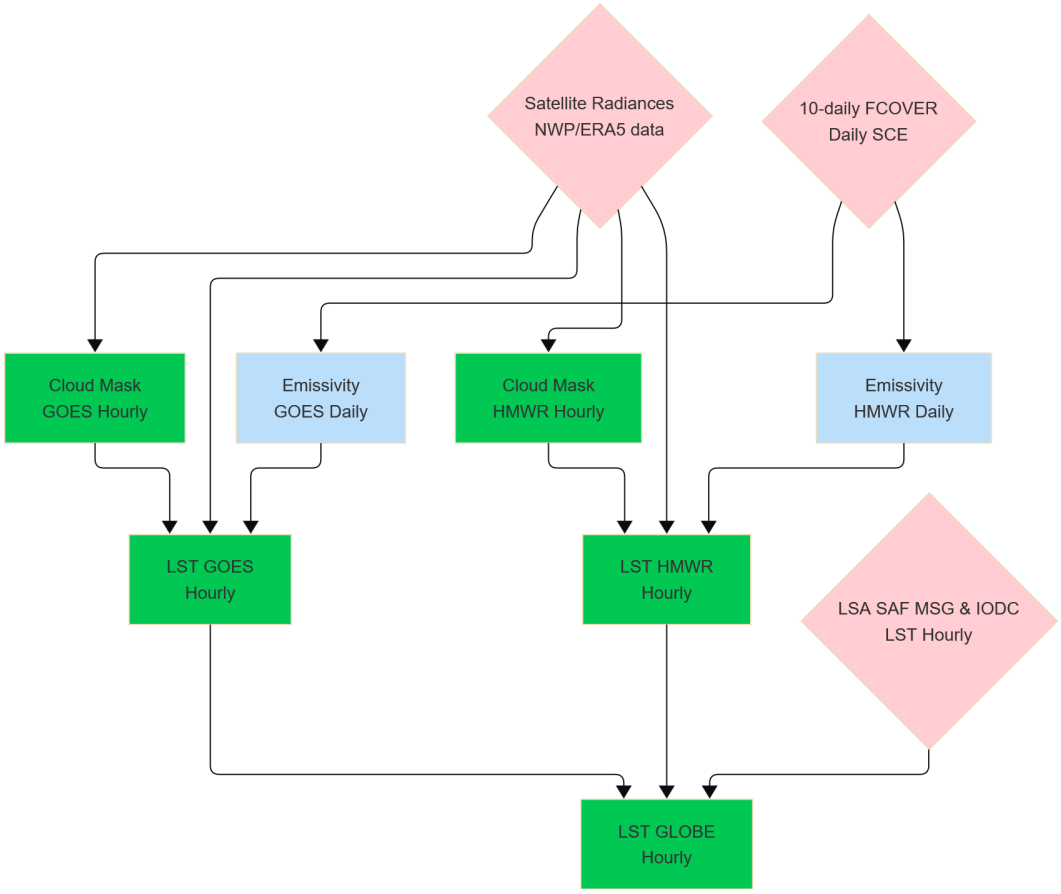


Figure 2: Flow diagram of the generation of LST GLOBE product.

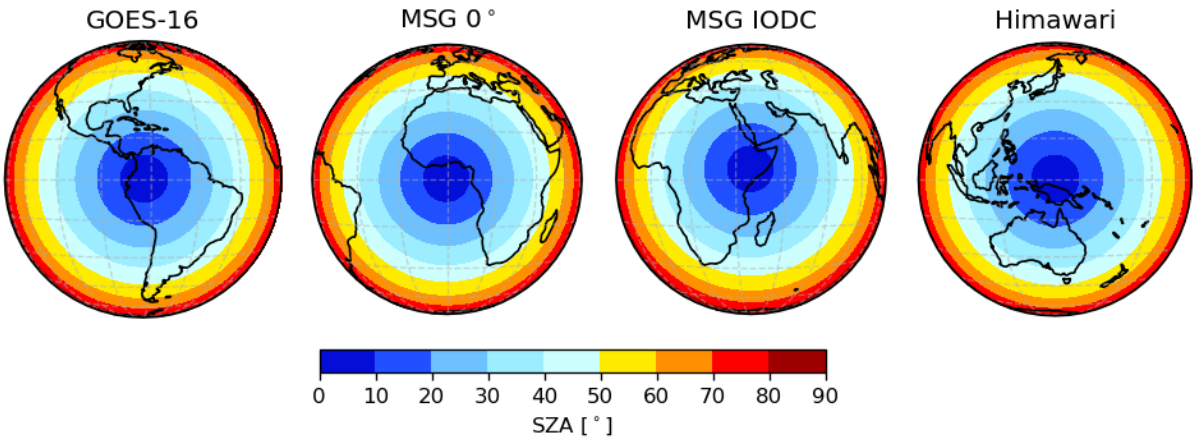


Figure 3: Satellite / View Zenith Angles for the four missions.

© CGLOPS Lot1 consortium

2.4 KNOWN LIMITATIONS

Over regions where different sensors overlap, users need to take into account that some differences exist in the way the LSA-SAF SEVIRI/MSG disk is processed, with respect to the processing of GOES and Himawari:

1) The retrieval algorithm calibration methodology: SEVIRI/MSG uses the methodology described in Freitas et al. (2010), while GOES and Himawari were calibrated using the methodology proposed by Martins et al. (2016), which used a more comprehensive choice of meteorological profiles [CGLOPS1_ATBD_LST-V3.0].
2) Although the methodology to estimate bare soil and vegetation emissivities for each land cover type is similar, the weight given to each component differs, since MSG/SEVIRI uses the LSA-SAF Fraction of Vegetation Cover product and both GOES and Himawari values obtained from the CLMS FCOVER products.

The vegetation cover method is widely used to estimate surface emissivity in TIR bands for LST retrievals, assuming representative bare-soil and vegetation emissivities for each land-cover class, though this assumption often fails in heterogeneous areas such as croplands, urban zones, or barren surfaces. To address this, the algorithm incorporates emissivity variability by assigning uncertainties to bare-soil and vegetation emissivities, derived from the maximum deviations found in the spectral emissivity database for each class, thus providing an upper-bound estimate of emissivity-related uncertainty that improves with database completeness. Emissivity uncertainties are generally below 1%, except for Snow/Ice and Barren classes, where they may reach 3%, while vegetation-cover effects contribute about 3% across all classes; accounting for these effects significantly reduces emissivity-driven uncertainty in vegetated areas. Ongoing research on improving bare-soil emissivity estimation may inform future updates of the CLMS LST product once sufficiently mature.

2.5 DIFFERENCES WITH PREVIOUS VERSION

To guarantee traceability, Table 11 in Annex 2: Differences with previous versions, summarizes the differences between successive product versions in terms of the 1) difference in calibration methodologies for individual sensors; 2) estimation of surface emissivity; 3) choice of the individual time slots for each sensor and 4) change in spatial coverage.

© CGLOPS Lot1 consortium

3. PRODUCT DESCRIPTION

The LST V3.0 product consists of instantaneous hourly LST fields for the areas covered by the geostationary satellites (see section 3.4.2).

3.1 FILE NAMING

For each time slot, the hourly LST products are named as follows:

c_gls_LST_<YYYYMMDDHHmm>_GLOBE_GEO_V<Major.Minor.Run>.nc

where

- <YYYYMMDDHHmm> gives the nominal time slot of the product. YYYY, MM, DD, HH, and mm denote the year, the month, the day, the hour, and the minutes, respectively;
- GLOBE indicates the global geographical extent of the product;
- GEO indicates that the product is based on GEOstationary satellite data;
- V<Major.Minor.Run> shows the processing line version used to generate these LST products. The updating of the Major and Minor version and Run number, as well as recommended usage, are described in Table 2.

Table 2: Explanation of the version numbering scheme and recommendations for handling version updates

Versions	Differences	Recommendations
Major	Significant change to the algorithm.	Do not mix various major versions in the same applications, unless it is otherwise stated.
Minor	Minor changes in the algorithm	Can be mixed in the same applications, but require attention or modest modifications
Run	Fixes to bugs and minor issues.	Later runs are drop-in replacements of former ones.

3.2 FILE FORMAT

The LST products are delivered as an internally compressed, hierarchical Network Common Data Form version 4 (netCDF4) data file, with metadata attributes compliant with version 1.6 of the Climate & Forecast conventions (CF V1.6 - http://cfconventions.org/)

© CGLOPS Lot1 consortium

3.3 PRODUCT CONTENT

The main netCDF4 file of each LST product contains the following data variables:

- **LST**: Land Surface Temperature
- **Q_FLAGS**: Bitwise quality flags providing information about the satellite(s) used to retrieve the data as well as the cloudiness of each grid point.
- **ERRORBAR_LST**: Error bar for LST
- **TIME_DELTA**: Observation time expressed in minutes between the observation time and the reference time that is written in the output file name.

To convert the digitally stored integer values of the data variables into their physical (often floating point) values, the user may need to multiply the integer values by the Scale Factor and then add the Additional Offset of Table 3. The scale factor and offset are stored in the *scale_factor* and *add_offset* metadata attributes in the netCDF4 files. Modern netCDF tools (e.g. netCDF4 python library) can thus apply the scaling and offset automatically when reading the data variables. The Q_FLAGS layer meaning area described in Table 4.

Table 3: Value encoding for the data variables

Variable	Physical values	Unit	Max digital value	Scale Factor	Additional Offset	Flagged values
LST	[-203.15, 353.15] K corresponding to -70 to 80°C	K	8000	1/100	273.15	-9999
ERRORBAR_LST	[0, 50] K	K	5000	1/100		-9999
Q_FLAGS	[0, 55]	-	55			-9999
TIME_DELTA	[-30, 30]	minute	30			-9999

Table 4: LST Quality Flags

Decimal value	Sat	Description
0	-	No data, Pixel out of Disk k
5	-	Sea
10	MSG (0° service)	Processed pixel, clear sky
15		Unprocessed pixel, cloud conditions
20	MSG IODC service	Processed pixel, clear sky
25		Unprocessed pixel, cloud conditions
30	Himawari	Processed pixel, clear sky
35		Unprocessed pixel, cloud conditions
40	GOES	Processed pixel, clear sky
45		Unprocessed pixel, cloud conditions
50	MULTI-MISSION	Processed pixel, clear sky
55		Unprocessed pixel, cloud conditions

As per the CF conventions, the netCDF files furthermore contain three dimensions and coordinated variables (**lat, lon** and **time**) and a variable for the grid mapping or coordinate reference system (**crs**).

The largely self-describing netCDF files contain a number of metadata attributes:

- on the file-level (Table 5)
- on the level of the data variables, with an example provided for the main 'LST' variable (Table 6)
- at the level of the coordinate variables for latitude and longitude, that provided as 1-D arrays of 8-byte floating point (Table 7);
- at the level of the coordinate variable for time dimension, that holds one value for the reference or nominal time slot (Table 8)
- at the level of the grid mapping variable (Table 9).

© CGLOPS Lot1 consortium

Table 5: File-level metadata attributes of LST products.

Attribute	Description	Possible Values
Conventions	Version of the CF-Conventions used	CF-1.6
algorithm_version	Version of the algorithms used to produce the global product.	MSG3-LST_8.0.2,HIMAWARI9-LST_4.2.0,GOES19-LST_4.2.0,MSG2-LST_7.15.2,LST_GLOBE_V3.0
contacts	Contact point(s) for questions or feedback on the LST products.	https://land.copernicus.eu/en/contact-service-helpdesk
copyright	Information on copyright	Copernicus Service Information 2025
history	Provides an audit trail for modifications to the original data netCDF file.	2026-03-26T07:35:43Z: Product generation
Identifier	Unique identifier for the product.	urn:cgls:global:lst_V3_0.025degree:LST_202603260700_GLOBE_GEO_V3.0.1
institution	Specifies the name of the institution that produced the product	IPMA
long_name	Extended product name	Land Surface Temperature
orbit_type	Orbit type of the satellites	GEO
parent_identifier	Identifier of the metadata series for the product in Copernicus Global Land metadata catalogue.	urn:cgls:global:lst_V3_0.025degree
platform	Names of the families (generations) of orbiting platforms	METEOSAT9,METEOSAT10,GOES19,HIMAWARI9,
processing_level	Product processing level	L4
processing_mode	Processing mode used when generating the product, with values Near Real Time, Offline, Reprocessing	Near Real Time
product_version	Version of the product	V3.0.1
references	Published or web-based references with further information about the product.	https://land.copernicus.eu/en/products/temperature-and-reflectance?tab=land_surface_temperature
sensor	Names of the sensors used	SEVIRI,SEVIRI,ABI,AHI,
source	The method of production (e.g. model, observations) of the original data on which the LST product is based.	Derived from EO satellite imagery

© CGLOPS Lot1 consortium

Attribute	Description	Possible Values
time_coverage_end	End date and time in UTC of the observation window used, formatted as per ISO 8601.	2026-03-26T07:15:00Z
time_coverage_start	Start date and time in UTC of the observation window used for the product, formatted as per ISO 8601.	2026-03-26T07:00:00Z
title	A brief description of the file content	Hourly Land Surface Temperature: GLOBE 2026-03-26T07:00:00Z

Table 6: Metadata attributes for the main LST data variable

Attribute	Description	Example value
_FillValue	Value used to represent missing or undefined data	-9999
add_offset	Offset to be added to the variable's contents in order to obtain the real (physical) values	273.15
ancillary_variables	Identification of other closely related variables in the same file	Q_FLAGS, ERRORBAR_LST, TIME_DELTA
cell_methods	Methods used to derive the data for each grid cell	area:mean where land
coverage_content_type	The coverage content type is an ISO 19115 code list describing the general content type of the resource.	physicalMeasurement
grid_mapping	Reference to the Coordinate Reference System variable	crs
long_name	A descriptive name that indicates a variable's content.	Land Surface Temperature
scale_factor	Multiplication factor for the variable's contents that must be applied in order to obtain the real (physical) values	0.01
standard_name	A standardised name from the CF Convention's standard name table that describes the variable's content (if available).	surface_temperature
units	Units of the variable's content	K
valid_range	Smallest and largest values for the variable	-7000, 8000

© CGLOPS Lot1 consortium

Table 7: Description of netCDF attributes for latitude and longitude coordinate variables

Attribute	Description	Example
axis	Identifies latitude, longitude (X), latitude (Y), vertical (Z), or time (T) axes.	X
long_name	A descriptive name that indicates a variable's content.	longitude
standard_name	A standardised name from the CF Convention's standard name table that describes the variable's content.	longitude
units	Units of the variable's content	degrees_east

Table 8: Description of netCDF attributes for time coordinate variable

Attribute	Description	Example
axis	Identifies latitude, longitude, vertical, or time axes.	T
calendar	Specific calendar assigned to the time variable. "standard" or "gregorian" refers to the default mixed Gregorian/Julian calendar.	standard
long_name	A descriptive name that indicates a variable's content.	time
units	Units of the variable's content, taken from Unidata's udunits library.	seconds since 2026-03-26 07:00:00

Table 9: Description of netCDF attributes for the grid mapping variable

Attribute	Description	Example
grid_mapping_name	Name used to identify the grid mapping or coordinate reference system	latitude_longitude
inverse_flattening	Used to specify the inverse flattening (1/f) of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	298. 257223563
longitude_of_prime_meridian	Specifies the longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum	0.0
semi_major_axis	Specifies the length, in metres, of the semi-major axis of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	6378137.0

© CGLOPS Lot1 consortium

3.4 PRODUCT CHARACTERISTICS

3.4.1 Projection & Grid Information

The product is displayed in a regular latitude/longitude grid with the ellipsoid WGS 1984 (Terrestrial radius=6378km). This coordinate reference system corresponds to EPSG code 4326. Following CF conventions, this coordinate reference system information is contained within the netCDF4 product file, in the grid_mapping variable called 'crs'. In addition, the coordinate variables for latitude and longitude (1-D arrays) and time (holding 1 value) define the coordinates of each grid cell.

The product resolution is 0.025° (approximately 3km at the equator) and is provided for all land surface pixels from longitude 180°E to 180°W and latitude 70°N to 70°S, resulting in a grid of 14400 columns and 5601 lines. Figure 4 shows the pixel average resolution as function of latitude for the geostationary platforms used and for the version 2 and version 3 LST product.

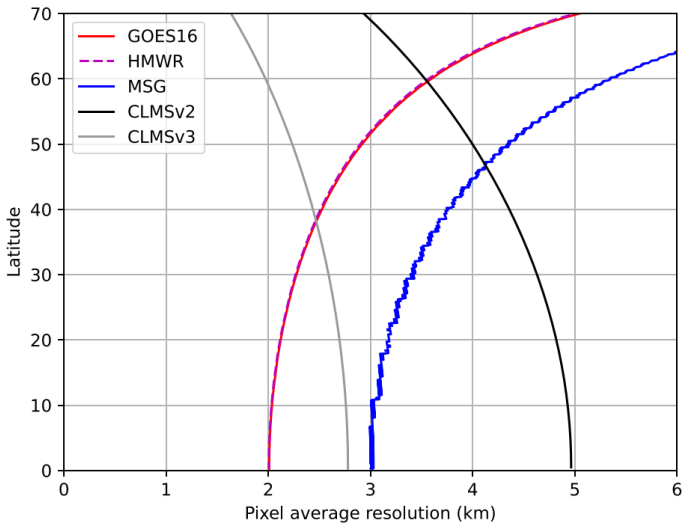


Figure 4 : Pixel average resolution as function of latitude for GOES16 (red), HMWR (dashed magenta), MSG (blue), CLMS LST previous version 2 (black) and CLMS LST version 3 (grey). For the Geostationary, it is assumed the central longitude of each platform.

3.4.2 Temporal Information

The LST product consists of instantaneous fields estimated every hour (00, 01, 02, 03,…, 23 UTC) using geostationary (GEO) satellite observations representative of such time steps. Not all the pixels in the image are observed at the same time due to differences in the satellite scan.

The duration of a full disk scan ranges from 12 min for MSG, to 10 min for the GOES-16 and HIMAWARI (20 min for their predecessors). The start time and the direction of the scan also

differ among GEO satellites. Figure 5 illustrates the direction and duration of scanning of each GEO satellite for a determined time slot.

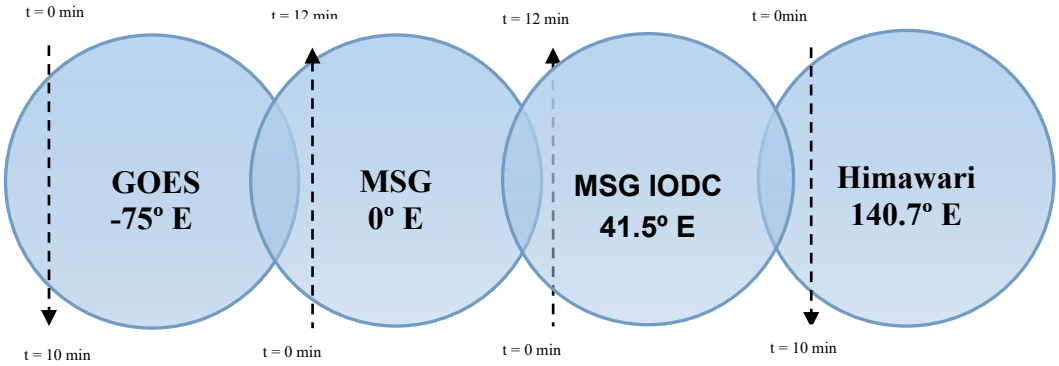


Figure 5: Illustration of the direction and duration of scanning of GEO satellites for the *t* UTC data.

Please note that, for a specific time slot, the LST product may be retrieved by using observations of about ±12 min apart from the reference UTC time *t* that is written in the file name (the nominal time slot). The difference between the nominal time slot and the actual pixel observation time is provided in the TIME_DELTA data variable.

3.5 Data Policies

The Copernicus component is governed by Regulation (EU) No 2021/696 of the European Parliament and of the Council of 28 April 2021 establishing the Union Space Programme and the European Union Agency for the Space Programme.

The Copernicus land monitoring products and services are made available on a principle of **full, open and free access**, as established by the Commission Delegated Regulation (EU) No 1159/2013 of 12 July 2013.

How to cite and reference the products and services of the Copernicus Land Monitoring Service is explained here: https://land.copernicus.eu/en/data-policy.

3.6 Access and Contacts

The LST products are available through the Copernicus Land Monitoring Service website at the address: https://land.copernicus.eu/en/products/temperature-and-

reflectance?tab=land_surface_temperature with contact information :
https://land.copernicus.eu/en/contact-service-helpdesk.

© CGLOPS Lot1 consortium

4. VALIDATION

The impact of changes in the LST processing as well as the overall quality of the LST V3.0 product have been assessed in the validation report [CGLOPS1_VR_LST-LST10-V3.0]. A summary is presented below.

4.1 PROCEDURE

An extensive quality assessment was performed to evaluate the performance of the CLMS LST V3.0 hourly over the study period from January 2018 to December 2025. The analyses included direct comparisons with in-situ observations from the Copernicus Ground-Based Observations for Validation (GBOV) service and performance assessments using other independent global LST datasets, together with cross-comparisons against the previous CLMS LST version. The validation against GBOV in-situ measurements provided a quantitative assessment of the CLMS LST product's accuracy and precision, while the comparison with other independent LST datasets (ERA5 reanalysis and Sentinel-3A SLSTR-A LST) allowed an evaluation more focused on the temporal stability and overall compliance with the CLMS LST product quality requirements.

4.2 MAIN RESULTS

Results in the Validation Report [CGLOPS1_VR_LST-LST10-V3.0] demonstrate a clear consistency of the product during the reprocessing period of eight years. The in-situ hourly validation (2018–2024) indicates that V3.0 has an overall bias of -0.43 K and absolute residuals of 1.76 K, with a root-mean-squared difference of 3.24 K. Conformity with the product's requirements show that 53 % of V3.0's satellite/in-situ matchups meet the threshold requirement (i.e. absolute error below or equal to 2 K) while 29 % meet the goal requirement (i.e. absolute error below or equal to 1 K).

Temporal stability was assessed by comparing V3.0 against ERA5 SKT (2018–2025) using anomaly-based monthly mean differences. These indicate reduced global discrepancies and enhanced stability with V3.0, showing systematically lower anomaly deviations and weaker trends relative to the previous CLMS LST version 2, reflecting enhanced temporal consistency, which is particularly evident before 2021. The stability of the reprocessed dataset was also assessed using independent satellite LST estimates from the SLSTR-A. The anomaly-based analysis highlights a gradual reduction of daytime differences (-0.45 K per decade) and small positive trend (+0.11 K per decade) for the nighttime period.

© CGLOPS Lot1 consortium

5. PRODUCT USAGE

5.1 TYPICAL USE

The CLMS LST product provides global, high-frequency observations of land surface temperature derived from a constellation of geostationary satellites, enabling consistent monitoring of surface thermal conditions.

The product is distributed as gridded NetCDF files that can be visualized using standard geospatial software. LST values are obtained by applying the provided scale factor and offset; while missing or low-quality observations (e.g. due to cloud contamination) are identified through dedicated quality flags. Each file represents LST conditions at given reference time, with additional information on acquisition timing and uncertainty provided through ancillary variables.

Standard applications of the product consist mainly in visualization and spatio-temporal analysis. LST fields can be displayed as maps or time-series to identify spatial patterns, seasonal variability, and diurnal temperature variations.

5.2 ADVANCED USAGE

Land surface processes and agriculture

LST is an essential component of the surface energy budget, controlling surface-emitted longwave radiation and enabling estimates of sensible and latent heat fluxes (Trigo et al., 2008). Its response to surface moisture conditions makes it a key variable for land-atmosphere interaction studies.

Under water-limited conditions, LST variations reflect soil moisture (SM) deficits, with diurnal temperature ranges strongly linked to SM variability (Trigo et al., 2015; Anderson et al., 2007). This relationship supports evapotranspiration and SM retrieval methods based on thermal observations (e.g., Li et al., 2021; Ghahremanloo et al., 2019; Tang et al., 2011).

In agricultural applications, geostationary LST captures the diurnal thermal dynamics associated with vegetation water stress, supporting irrigation management and agricultural drought monitoring (e.g., Muro et al., 2018), as well as the development of LST-based drought indices using instantaneous or time-series approaches (e.g., Rhee et al., 2010; Moran et al., 1994).

5.2.1 Hydrology and water cycle

LST is widely used in water resource management and hydrological applications due to its sensitivity to surface energy partitioning and land–atmosphere coupling processes (Trigo et al.,

2015; Anderson et al., 2012). It provides indirect information on surface moisture conditions and hydrological variability, supporting the analysis of the surface water balance and drought-related anomalies (Stoyanova et al., 2019). In addition, atmospheric water vapour content is a key factor affecting LST retrieval accuracy (Freitas et al., 2010). These characteristics support the use of LST in hydrological monitoring and the characterization of land surface water stress at regional and global scales.

Urban Climate

The surface urban heat island refers to the phenomenon in which urban land surfaces exhibit higher temperatures than surrounding rural areas, and is a key indicator of the urban thermal environment (Oke, 1982). Given its suitability for capturing urban heat island diurnal cycles and intra-day variability across cities, urban climate has been extensively studied using LST datasets derived from different satellite platforms and diverse methodologies (e.g., Morgese et al., 2025; Gorai et al., 2024; Naserikia et al., 2023; Elmes et al., 2020; Zhao et al., 2014; Cui & Foy, 2012), providing insight into the temporal evolution and structural characteristics of urban thermal environments.

Thermal anomaly monitoring

LST is suitable for detecting heatwaves and other extreme thermal events by identifying deviations from expected spatial or temporal temperature patterns (e.g., Kim et al., 2025; Martins et al., 2024). Thermal anomalies correspond to positive or negative departures from a reference LST state and are commonly identified using statistical thresholds applied to spatial, temporal, or combined climatological baselines (Li et al., 2023). These approaches enable the detection of anomalous surface temperature conditions, supporting the identification of localized events as well as large-scale, persistent thermal extremes such as heatwaves, relevant for climate and environmental monitoring.

Climate studies

LST is increasingly applied in climate studies to monitor variability and support the derivation of climate change indices (Li et al., 2023). Small shifts in mean surface temperature can lead to disproportionate changes in extremes, making thermal extremes particularly relevant for climate studies (e.g., Reiners et al., 2023; Caloiero, 2017). In this context, LST observations provide valuable complementary information to air temperature datasets, especially in regions with limited in-situ coverage or coarse-resolution reanalysis products.

Satellite–derived LST can be used to estimate climate indices related to temperature extremes, including intensity (e.g. maximum and minimum LST) and range metrics such as the diurnal temperature range. These indicators support the analysis of climate variability, long-term trends, and changes in extreme thermal conditions from regional to global scales (e.g., Alexander et al., 2006).

Model-related applications

LST is widely used in model evaluation (e.g., Ghent et al., 2024; Martin et al., 2019) due to its spatial coverage and sensitivity to land–atmosphere processes, enabling validation and intercomparison of land surface and climate models, particularly in the assessment of surface energy balance and temperature-related processes. Evaluation typically relies on comparisons with in-situ observations using standard performance metrics (e.g., bias, RMSE, MAE, coefficient of determination).

In addition, satellite–derived LST is increasingly assimilated into numerical models to constrain surface states and improve the representation of land–atmosphere interactions. It also contributes to reanalysis systems by providing observational constraints that enhance the consistency and accuracy of long-term climate datasets.

Beyond evaluation and assimilation, LST is also used in data-driven modelling approaches, including machine learning methods, to represent and predict surface temperature dynamics (e.g., Hoang et al., 2025; Pandle et al., 2024). These approaches support applications such as urban climate analysis and environmental monitoring.

6. REFERENCES

Alexander, L. V., et al. (2006). Global observed changes in daily climate extremes of temperature and precipitation, J. Geophys. Res., 111, D05109. https://doi.org/10.1029/2005JD006290

Anderson, M. C., Allen, R. G., Morse, A., & Kustas, W. P. (2012). Use of Landsat thermal imagery in monitoring evapotranspiration and managing water resources. *Remote Sensing of Environment*, *122*, 50–65. https://doi.org/10.1016/j.rse.2011.08.025

Anderson, M. C., J. M. Norman, J. R. Mecikalski, J. A. Otkin, & Kustas, W. P. (2007). A climatological study of evapotranspiration and moisture stress across the continental United States based on thermal remote sensing: 1. Model formulation, J. Geophys. Res., 112, D10117. https://doi.org/10.1029/2006JD007506

Caloiero, T. (2017). Trend of monthly temperature and daily extreme temperature during 1951–2012 in New Zealand. *Theoretical and Applied Climatology*, *129*(1), 111–127. https://doi.org/10.1007/s00704-016-1764-3

Cui, Y. Y., & B. de Foy (2012). Seasonal Variations of the Urban Heat Island at the Surface and the Near-Surface and Reductions due to Urban Vegetation in Mexico City. *J. Appl. Meteor. Climatol.*, 51, 855–868. https://doi.org/10.1175/JAMC-D-11-0104.1

Elmes, A., Healy, M., Geron, N., Andrews, M. M., Rogan, J., Martin, D. G., Sangermano, F., Williams, C. A., & Weil, B. (2020). Mapping spatiotemporal variability of the urban heat island across an urban gradient in Worcester, Massachusetts using in-situ Thermochrons and Landsat-8 Thermal Infrared Sensor (TIRS) data. *GIScience & Remote Sensing*, *57*(7), 845–864. https://doi.org/10.1080/15481603.2020.1818950

Ghahremanloo, M., Mobasheri, M. R., & Amani, M. (2019). Soil moisture estimation using land surface temperature and soil temperature at 5 cm depth. *International Journal of Remote Sensing*, *40*(1), 104–117. https://doi.org/10.1080/01431161.2018.1501167

Ghent, D., Anand, J. S., Veal, K., & Remedios, J. (2024). The Operational and Climate Land Surface Temperature Products from the Sea and Land Surface Temperature Radiometers on Sentinel-3A and 3B. Remote Sens., 16, 3403. https://doi.org/10.3390/rs16183403

Gorai, N., Bandyopadhyay, J., Halder, B., Ahmed, M. F., Molla, A. H., & Lei, T. M. T. (2024). Spatio-Temporal Variation in Landforms and Surface Urban Heat Island in Riverine Megacity. *Sustainability*, *16*(8), 3383. https://doi.org/10.3390/su16083383

Hoang, N.-D., Tran, V.-D., & Huynh, T.-C. (2025). From Data to Insights: Modeling Urban Land Surface Temperature Using Geospatial Analysis and Interpretable Machine Learning. Sensors, 25, 1169. https://doi.org/10.3390/s25041169

Kim, W., M., Simpson, I. R., Terray, L., & Collazo, S. (2025). The Unprecedented Late-Summer 2023 Heatwave in Southeastern South America: Attribution and future projection of similar events. Weather and Climate Extremes 48, 100772. https://doi.org/10.1016/j.wace.2025.100772

Kogan F, Salazar L, Roytman L. 2012. Forecasting crop production using satellite-based vegetation health indices in Kansas, USA. *International Journal of Remote Sensing*. https://doi.org/10.1080/01431161.2011.621464

Kogan FN. 2001. Operational space technology for global vegetation assessment. *Bulletin of the American Meteorological Society*. https://doi.org/10.1175/1520-0477(2001)082<1949:OSTFGV>2.3.CO;2

Li, Z.-L., Leng, P., Zhou, C., Chen, K.-S., Zhou, F.-C., & Shang, G.-F. (2021). Soil moisture retrieval from remote sensing measurements: Current knowledge and directions for the future. *Earth-Science Reviews*, 218, 103673. https://doi.org/10.1016/j.earscirev.2021.103673

Li, Z.-L., Wu, H., Duan, S.-B., Zhao, W., Ren, H., Liu, X., et al. (2023). Satellite remote sensing of global land surface temperature: Definition, methods, products, and applications. Reviews of Geophysics, 61, e2022RG000777. https://doi.org/10.1029/2022RG000777

Martin, M. A., Ghent, D., Pires, A. C., Göttsche, F. M., Cermak, J., & Remedios, J. J. (2019). Comprehensive in situ validation of five satellite land surface temperature data sets over multiple stations and years. *Remote Sensing*, *11*(5). https://doi.org/10.3390/rs11050479

Martins, J. P. A., Caetano, S., Pereira, C., Dutra, E., & Cardoso, R. M. (2024). A satellite view of the exceptionally warm summer 0f 2022 over Europe. Nat. Hazards Earth Syst. Sci., 24, 1501–1520. https://doi.org/10.5194/nhess-24-1501-2024

Moran, M. S., Clarke, T. R., Inoue, Y., & Vidal, A. (1994). Estimating crop water deficit using the relation between surface-air temperature and spectral vegetation index. *Remote Sensing of Environment*, 49(3), 246–263. https://doi.org/10.1016/0034-4257(94)90020-5

Morgese, S., Zhang, W., Casale, F., & Bocchiola, D. (2025). A Hydrological–Energy Balance Model to Assess Land Surface Temperature at the Urban Scale: The Case Study of Milano, Italy. *Meteorological Applications* 32, no. 3: e70069. https://doi.org/10.1002/met.70069

Muro, J., Strauch, A., Heinemann, S., Steinbach, S., Thonfeld, F., Waske, B., Diekkrüger, & B. (2018). Land surface temperature trends as indicator of land use changes in wetlands, International Journal of Applied Earth Observation and Geoinformation, Volume 70, 62-71, ISSN 1569-8432. https://doi.org/10.1016/j.jag.2018.02.002

Naserikia, M., Hart, M. A., Nazarian, N., Bechtel, B., Lipson, M., & Nice, K.A. (2023). Land surface and air temperature dynamics: The role of urban form and seasonality, Science of

The Total Environment, Volume 905, 167306, ISSN 0048-9697. https://doi.org/10.1016/j.scitotenv.2023.167306

Oke, T. R. (1982). The energetic basis of the urban heat island. *Quarterly Journal of the Royal Meteorological Society*, 108(455), 1–24. https://doi.org/10.1002/qj.49710845502

Pandle, C. B., Egbueri, J. C., Costache, R., Sidek, L. M., Wang, Q., Alshehri, F., Din, N. M., Gautam, V. K., & Pal, S. C. (2024). Predictive modeling of land surface temperature (LST) based on Landsat-8 satellite data and machine learning models for sustainable development, Journal of Cleaner Production, Volume 444, 141035, ISSN 0959-6526. https://doi.org/10.1016/j.jclepro.2024.141035

Reiners, P., Sobrino, J., & Kuenzer, C. (2023). Satellite-Derived Land Surface Temperature Dynamics in the Context of Global Change—A Review. Remote Sens., 15, 1857. https://doi.org/10.3390/rs15071857

Rhee, J., Im, J., & Carbone, G. J. (2010). Monitoring agricultural drought for arid and humid regions using multi-sensor remote sensing data. *Remote Sensing of Environment*, 114(12), 2875–2887. https://doi.org/10.1016/j.rse.2010.07.005

Stoyanova, J., Georgiev, C., Neytchev, P., & Kulishev, A. (2019). Spatial-Temporal Variability of Land Surface Dry Anomalies in Climatic Aspect: Biogeophysical Insight by Meteosat Observations and SVAT Modeling. Atmosphere, 10, 636. https://doi.org/10.3390/atmos10100636

Tang, R. L., Li, Z. L., & Chen, K. S. (2011). Validating MODIS-derived land surface evapotranspiration with in situ measurements at two AmeriFlux sites in a semiarid region. *Journal of Geophysical Research*, 116(D4), D04106. https://doi.org/10.1029/2010JD014543

Trigo, I. F., Boussetta, S., Viterbo, P., Balsamo, G., Beljaars, A., & Sandu, I. (2015). Comparison of model land skin temperature with remotely sensed estimates and assessment of surface-atmosphere coupling. *Journal of Geophysical Research*, *120*(23), 12,096-12,111. https://doi.org/10.1002/2015JD023812

Trigo, I. F., Monteiro, I. T., Olesen, F., & Kabsch, E. (2008). An assessment of remotely sensed land surface temperature. *Journal of Geophysical Research Atmospheres*, *113*(17). https://doi.org/10.1029/2008JD010035

Zhao, L., Lee, X., Smith, R. B., & Oleson, K. (2014). Strong contributions of local background climate to urban heat islands. *Nature*, 511(7508), 216–219. https://doi.org/10.1038/nature13462

7. REFERENCES

Freitas, S. C., I. F. Trigo, and J. M. Bioucas-Dias, F. Göttsche, 2010: Quantifying the Uncertainty of Land Surface Temperature Retrievals from SEVIRI/Meteosat. *IEEE Trans. Geosci. Remote Sens*., 48, https://doi.org/10.1109/TGRS.2009.2027697

Martins, J.P., Trigo, I., Bento, V., da Camara, C., 2016: "A Physically Constrained Calibration Database for Land Surface Temperature Using Infrared Retrieval Algorithms". *Remote Sens*. 8(10), 808. https://doi.org/10.3390/rs8100808

Trigo, I. F., I. T. Monteiro, F. Olesen, and E. Kabsch, 2008a, "An assessment of remotely sensed Land Surface Temperature", *J. Geophys. Res*., vol 113, D17108, https://doi.org/10.1029/2008JD010035.

Trigo, I.F., Peres, L.F., DaCamara, C.C., Freitas, S.C., 2008b. Thermal land surface emissivity retrieved from SEVIRI/Meteosat. IEEE Trans. Geosci. Remote Sens. 46, 307–315. https://doi.org/10.1109/TGRS.2007.905197

8. ANNEX 1: REQUIREMENTS

According to the applicable document [AD1], specific technical details and requirements are described below (Table 10).

8.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

Table 10: CGLOPS LST product specifications.

LST – LAND SURFACE TEMPERATURE	
The Land Surface Temperature (LST) is the radiative skin temperature of the land surface, as measured in the direction of the remote sensor [K].	
Geometric Properties:	
Spatial resolution	5 km
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage	Global
Time span	Hourly and 10-day period
Timeliness	Hourly: within 4 hours; within 2 days (optimally 1 day) after the end of 10-day period
Uncertainty (2-sigma)	Threshold: 2 K, Goal: 1 K
Stability (per decade)	Threshold: 0.3 K, Goal: 0.1 K

8.2 FURTHER REQUIREMENTS

8.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

8.2.2 Data structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, and conversion to physical values formulas.

Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product;
- The per-pixel date of the individual measurement or the start-end dates of the period covered;
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result;

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

8.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets shall be available in NETCDF. Additional format, such as COG or ZAR, format can be proposed for production or requested by the contracting authority during the Framework.

8.2.4 Uncertainties and validation

Uncertainties indicated in the product specifications (Table 10) follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate, the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL datasets and/or Ground-Based Observations for Validation (GBOV, https://land.copernicus.eu/en/products/GBOV) of Copernicus Global Land component when biophysical parameters are available.

8.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu;

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary datasets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

8.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination. The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products

[2] https://land.copernicus.eu/en/products/global-image-mosaic

9. ANNEX 2: DIFFERENCES WITH PREVIOUS VERSIONS

Table 11: Differences between product versions.

Change	V1.2	V2.0	V2.1	V2.2	V3.0
Algorithm Calibration Coefficients	Selection of the atmospheric profiles used for algorithm calibration: • Follows Trigo et al. (2008a) for MSG; • Complete coverage of the $T_{skin}/TCWV$ phase space Martins et al., (2016) for GOES-16 and Himawari.	Selection of the atmospheric profiles used for algorithm calibration: • Follows Trigo et al. (2008a) for MSG 0º and IODC; • Complete coverage of the $T_{skin}/TCWV$ phase space Martins et al., (2016) for GOES-16 and Himawari. New optimized set of profiles.	Same as V2.0	Same as V2.0 / V2.1	Same as V2.2
Emissivity estimation	Static (except within the MSG disk), based on land cover information and the Vegetation Cover Method (Trigo et al., 2008b). The weighting between bare ground and vegetation emissivities uses a fixed value for each IGBP land cover type. For MSG, the	Dynamic, estimated using the Vegetation Cover Method (Trigo et al., 2008b). The weighting between bare ground and vegetation emissivities uses the dekadal CLMS 1 km FCOVER product. In the case of the MSG 0º and IODC missions, the weight is given by the LSA-SAF daily FVC product.	Relies on 300 m FCOVER V1.1. Uses the CGLOPS SCE 1 km NH V1.0 product to correct emissivity over pixels with snow.	Same as V2.0 / V2.1	Same as V2.2 with varying FCOVER and SNOW data inputs (see annex 1 of ATDB)

© CGLOPS Lot1 consortium

	weight is given by the LSA-SAF daily FVC product.				
Individual time slots for each sensor	MSG: T-15 min GOES-16: T = 0 min HMWR: T = 0 min	MSG: T = 0 min GOES-16: T = 0 min HMWR: T = 0 min	Same as V2.0	Same as V2.0 / V2.1	Same as V2.2
Spatial Coverage	MSG 0º, GOES-16 and HMWR-8	MSG 0º, MSG IODC, GOES-16 and HMWR-8	Same as V2.0	Same as V2.0 / V2.1	Same as V2.2
Others				Fix coordinate reference system (crs) inverse_flattening attribute and global attributes “sensor”,platform” No change in the algorithm or in the actual products content.	Same as V2.2, and use of SAF NWC cloud mask for GOES and HMWR
Temporal coverage	19/06/2010 to 18/01/2021	18/01/2021 to 31/12/2023	01/01/2023 to 02/02/2025	18/01/2021 to 2026	Reprocessed from 2018-01-01 to near real time

© CGLOPS Lot1 consortium